

EDS1/80901/24
ABDULMALIK KYENZI

a) Discuss the term constraints as used in databases and give at least THREE constraints (4 marks)
Constraints in databases are rules applied to table columns to enforce data integrity and accuracy. They ensure that the data entered into the database meets specific rules and guidelines.

Three common constraints:

PRIMARY KEY – Ensures each record in a table is unique and not null.

FOREIGN KEY – Enforces referential integrity by linking records in different tables.

NOT NULL – Ensures a column cannot have a null (empty) value.

b) State THREE conditions that a field should meet to qualify to be a primary key (3 marks)
Uniqueness – Every value in the field must be unique across all records.

Not Null – The field must contain a value; it cannot be left blank.

Immutable – The value should not change over time to maintain consistent identification.

c) State TWO differences between foreign keys and primary keys in a database (4 marks)

Primary Key	Foreign Key
-------------	-------------

Uniquely identifies a record in the same table.	Refers to the primary key in another table.
---	---

Cannot contain NULL values.	Can contain NULL values (if not mandatory).
-----------------------------	---

d) Discuss data definition language (DDL) in SQL and give three commands used (4 marks)

Data Definition Language (DDL) is a subset of SQL used to define and manage database structure and schema.

Three common DDL commands:

CREATE – Creates new tables, databases, views, etc.

ALTER – Modifies existing database structures.

DROP – Deletes tables, views, or databases.

e) Differentiate a composite key from a compound key as used in database (4 marks)

Composite Key: A key made up of two or more columns, where all columns together uniquely identify a record, and none of the individual columns can uniquely identify the record.

Compound Key: A key also made up of multiple columns, but at least one of the columns can uniquely identify the record on its own.

Note: In many contexts, these terms are used interchangeably, but technically, the above distinction applies in specific database theory contexts.

f) Use the table below to answer the following questions:

Assume this is the table provided:

EMPNO	ENAME	JOB	SALARY	DEPTNO
1001	Alice	Manager	70000	1010
1002	Bob	Analyst	50000	1010

1003 Charlie Clerk 30000 1020

g. Write SQL command to create the above table (4 marks)

sql

Copy

Edit

```
CREATE TABLE Employee (  
EMPNO INT PRIMARY KEY,  
ENAME VARCHAR(50),  
JOB VARCHAR(50),  
SALARY DECIMAL(10, 2),  
DEPTNO INT  
);
```

ii. Write SQL command to insert the first TWO rows of the above table (4 marks)

sql

Copy

Edit

```
INSERT INTO Employee (EMPNO, ENAME, JOB, SALARY, DEPTNO)  
VALUES (1001, 'Alice', 'Manager', 70000, 1010);
```

```
INSERT INTO Employee (EMPNO, ENAME, JOB, SALARY, DEPTNO)  
VALUES (1002, 'Bob', 'Analyst', 50000, 1010);
```

iii. Write SQL command to display the employees who belong to DEPTNO 1010 (2 marks)

sql

Copy

Edit

```
SELECT * FROM Employee  
WHERE DEPTNO = 1010;
```